Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Electrochemical theory of corrosion

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Electrochemical theory of corrosion
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Electrochemical theory of corrosion

1 Electrochemical theory of corrosion

Study focus: Electrochemical theory of corrosion. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will start with a new module today on corrosion chemistry.

Key details

- Today's class we will understand what is corrosion, the different types of corrosion, the electrochemical theory of corrosion, and finally we will look at corrosion of iron.
- Whenever the metal surface is exposed to the environment, the environment is going to cause this destruction of the metal.
- It could be water or it could be simply moisture in the air, acids, bases, salts, oils, aggressive metal polishes and other types of solid and liquid chemicals.
- And what is important for us to know is that it is a spontaneous process.
- means whenever a metal is exposed to the environment, the reaction is going to be spontaneous.
- We need not supply any energy for the process to take place.
- And that is the reason why the moment the metal is exposed to any of these chemicals it is undergoing corrosion.
- And most importantly, corrosion is a nuisance for us because it is not just the direct metal loss, it is the indirect losses which happen when corrosion takes place.
- There could be oil pipelines which would get burst.
- So all these are going to result in colossal losses, and therefore corrosion becomes very, very important study for us.
- What exactly happens during corrosion?
- We know that pure metal.
- Is getting converted into undesirable products.

- We made structures made of pure metal, but they are getting converted into undesirable products.
- Rusting of iron is something all of us have experienced.
- There is a reddish brown scale which is getting formed on the iron and steel objects.
- And what is this?
- This is actually hydrated ferric oxide.
- And many times in the when you have these copper articles, we see that they get a greenish.
- color on the surface.
- It is due to basic pupric carbonate.
- Another phenomenon which we obser observe and that is due to corrosion.
- It is getting converted into CUCO3 plus C U O H twice.
- which is actually covered with a thin copper skin and it was originally brown in colour as you can see in the picture.
- But with time due to exposure to the environment, this has resulted in the formation of the basic copper carbon.

Example or application from the lecture

- Firstly, let us define what is corrosion.
- For example, it could result in a bridge collapsing, or it could result in air crashes.
- Let us understand.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

Which we saw here, which is known as blue green patina, and as we see it today, it is this bluish green in color.

Key details

- Apart from this, we can also have silver getting tarnished when it is exposed to an H2S atmosphere.
- We know that silver particles they get this black color.
- that the metal is getting converted into a compound which is n not desirable for us.
- Why do metals undergo corrosion?
- Or rather, why is corrosion a spontaneous process?
- What is happening is in nature, most of the metals are in the form of their compounds.
- that is the mineral ores which are available to us, they are in the form of compounds like they could be oxides, sulphides, carbonates, chlorides, and many other compounds.
- Then when we do an extraction process.
- Extraction is always an endothermic process because we are either supplying heat or we are supplying some electrical energy.
- And because any system tries to go back to the lower energy state.
- That's why in nature we find that this process metal getting converted into the compound or any d undesirable product is.

- B is a spontaneous process because when you extracted the metal it you got it to a higher energy state by supplying energy, therefore there's a natural tendency for it to go back to the combined form.
- One is dry corrosion.
- The other is wet corrosion.
- In dry corrosion, the metal is being attacked directly by the gases.
- These gases are attacking metal surface.
- Throughout this chapter we have to.
- Understand that it only affects the surface of the metal, the bulk of the metal does not get affected by corrosion.
- Because the metal has an affinity for these gases.
- Say it could be oxygen.
- chlorine, H, ammonia, any of the gases SO2 which are present in the environment around, they are directly attacking the metal surface and therefore it is giving rise to some a product.
- on the metal surface.
- So, therefore, we say it's a direct chemical attack, and it is called dry corrosion.
- And when this happens, we say that it is a electrochemical attack on the metal.
- Anode and cathode formation at as we studied in galvanic cells.

Example or application from the lecture

- So let us answer the first question.
- Let us understand this.
- This metal has you have given this much energy, it's an endothermic process, so they end up at a higher energy state as compared to the ore from which they were formed.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

and cathode on the surface of the metal.

Key details

- We have to remember always that it is happening only on the surface.
- They are of course made of the anode and the cathode.
- What do we have?
- We have this metal.
- So basically, metal is getting converted into a ion that means into its compounds.
- That's how we define corrosion.
- That metal gets converted into an undesired compound.
- We see that the metal is getting corroded and it is going to get destroyed.
- Of course, there are n number of electrons which are getting generated in this process.
- Remember, it is the same metal, and one part of it is acting as the cathode.
- Now at the cathode we have to remember that metal can never undergo reduction because at the cathode what is the process which is supposed to take place?

- It is reduction process.
- The metal can never undergo reduction.
- Because it's already in the lowest reduced form.
- Therefore, it is the environment.
- It will never happen.
- So it is the environment because we know that again I would and cathode oxidation and reduction.
- Should happen simultaneously.
- We cannot have just oxidation happening and no reduction.
- Some species in the environment is going to undergo reduction and that will result in the formation of AN minus.
- Fact we have to remember that the anode part of the metal is getting corroded whereas the cathode part is completely unaffected by corrosion because the metal remains intact.
- It's only the environment which is taking part in the reduction process.
- These electrons are going to move towards the cathode and what would be the electronic conducting media?
- That would be the metal itself.
- And we have the AN minus which got formed at the cathode.
- Between there will be formation of corrosion product.
- These two are going to combine the Mn plus ions and the AN minus ions are going to move towards.
- each other and somewhere in between we will find the formation of the corrosion product.
- And remember why do we need a conducting medium for such a phenomena to happen?
- On the metal surface, whereas these ions need a medium to move, and therefore a conducting medium is what is required for corrosion to happen.
- And we should always remember that as long as the anode and the cathode reaction happens simultaneously.
- Corrosion process is going to continue.
- At the anode, we don't have any other reaction which can take place, but that the metal is getting.
- oxidized and therefore giving out n electrons and we have Mn plus ions which are getting formed.
- But at the cathode, since it is not the metal which is involved in the reaction, you could you are having the environment which is taking part in the reaction.

Example or application from the lecture

- Let us check out what are the reactions which are actually happening.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 15:00–19:59

To simplify, we can understand there can be two types of reaction, either liberation of hydrogen or the absorption of oxygen.

Key details

- What happens when there is liberation of hydrogen?
- And again the medium could be acidic or it could be neutral or alkaline.
- If it is acidic medium, we observe that obviously there will be lot of H plus ions and it's a cathodic reaction so electron will be taken up and we get liberation of hydrogen.
- In neutral or alkaline medium again reduction because we are at the cathode.
- And it is either a neutral or alkaline medium, then this is the reaction.
- If it's an acidic medium this will be the predominant reaction.
- Again we could have in the acidic medium or in the neutral or alkaline medium.
- Absorption of oxygen will obviously happen when there is lot of oxygen present in the atmosphere.
- If it is a neutral or alkyne medium, then again we have water, but now oxygen is getting.
- absorbed with four electrons taking part in the reaction and H minus.
- Add the anode we know that Fe2 plus ions are going to get formed and add the cathode.
- Most common reaction would be in a neutral medium oxygen is getting absorbed.
- Fe becoming Fe two plus two electron and most common would be of course if it's an acidic medium as in this case then we will find that water will be liberated, oxygen is being used and water is liberated.
- But most commonly we find neutral mediums.
- medium and we get 2 H2O plus O2 plus 4 electron is giving you 4 O H minus.
- Now what happens next?
- The Fe2 plus ions which are got which were formed here, and this is the conducting medium as can be seen here.
- This Fe2 plus ions and the OH minus ions move towards one another and we get the formation of FeOH twice.
- somewhere between the anode and the cathode regions.
- You can see here that the corrosion product is getting formed, of course it will be on the surface because corrosion is a surface phenomenon.
- Fe2 plus and OH minus are diffusing towards one another and we get FeOH twice, but we know that rust is not FeOH twice.
- So what happens next?
- Fe2 plus is going to get converted or oxidized into Fe three plus.
- We know that Fe three plus is a more stable state as compared to Fe two plus.
- Two plus has got converted to F three plus and that is yellow dust.
- That is because you provided an oxidizing atmosphere.
- And this is known as black rust.

Example or application from the lecture

- Let us see.
- Let us now look at a specific case and that is corrosion of iron.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 20:00–24:59

of oxygen FE3O4 if is actually Fe O.

Key details

- Fe two O three.
- It is known as ferrosopheric oxide, Fe three O4, and therefore in limited oxygen we will see the formation of black rust.
- So now in the next class we are going to talk about the different types of corrosion.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Electrochemical theory of corrosion transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Electrochemical theory of corrosion transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00019	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00021	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...

C Revision and retrieval practice